An Empathetic AI Driven Multilingual, Emotion-Aware Recruitment System

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An Empathetic AI Driven Multilingual, Emotion-Aware Recruitment System

Project Report submitted in the partial fulfilment

of

Bachelor of Technology

In

Computer Engineering

by

Sanjna Deva (B078)

Soumik Sahu (B168)

Under the supervision of

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(2022-2026)

CERTIFICATE

This is to certify that the project entitled **"An Empathetic AI-Driven Multilingual, Emotion-Aware Recruitment System"**, has been done by **Ms. Sanjna Deva** and **Mr. Soumik Sahu** under my guidance and supervision & has been submitted in partial fulfilment of the degree of Bachelor of Technology in Computer Engineering of MPSTME, SVKM's NMIMS (Deemed-to-be University), Mumbai, India.

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ABSTRACT

In recruitment, biases, inefficiencies, and inconsistencies often lead evaluators to misjudge candidates. The impact of these challenges extends to organizational recruitment systems and productivity. To tackle this issue, we present an automated, AI-enabled interview system formulated to be voice-centered to evaluate and objectively score interviews based on data. The envisioned systems apply cutting-edge technology to conduct a complete person assessment, including emotion assessment through RAVDESS and associated datasets, automated multilingual speech-to-text transcription through VAPI, and text response evaluation with contemporary transformer-based language models such as BERT and Gemini.

For fairness and objectivity, the product uses a rubric-based evaluative framework based on tier of AI, where each tier has a patched rubric to ensure AI rigor in scoring across language, emotion and context. The website is built with Next.js for compliant frontend, while Firebase is real-time database control, authentication and recruiter dashboard resources. The system directly addresses the clarifiability of AI operating within the system, ensuring that AI-based actions are transparent to users.

Table Of Contents

Topics	Page
List of Figures	i
List of Tables	xvi
Abbreviations	xix
Chapter 1 Introduction	
1.1 Background of Project Topic	
1.2 Statement of Motivation and Scope of Report	
1.3 Problem Statement	
1.4 Salient Contribution	
1.5 Organization of Report	
Chapter 2 Literature survey	
2.1 General Introduction to Topic	
2.2 Exhaustive Literature Survey	
Chapter 3 Methodology and Implementation	
3.1 Block Diagram	
3.2 Hardware Description	
3.3 Software Description and Flowchart	
3.4 Algorithm	
3.5 Summary	
Chapter 4 Result and Analysis	
4.1 System Evaluation Overview	

4.2 Fairness and Transparency in Assessment

4.3 Emotion Awareness and Candidate Behavior Insight

4.4 Semantic Understanding and Contextual Evaluation

4.5 Feedback Generation and Recruiter Experience

Chapter 5 Advantages, Limitations and Applications

5.1 Advantages

5.2 Limitations

5.3 Applications

Chapter 6 Conclusion and Future Scope

6.1 Conclusion

6. 2 Future Scope

References

Appendix A: Soft Code Flowcharts

Appendix B: Data Sheets

Appendix C: List of Components

Appendix D: List of Paper Presented and Published.

List of Figures

Fig No	Name of the figure	Page No
1	System Pipeline of Project	XXVII
2	AI powered hiring system detailed flowchart	XXXII

List of Tables

Table No	Name of the table	Page No
1	Comparison of Commercial Recruitment Platforms and Technologies	XVII
2	Component and their specifications	XXVII

List of Abbreviation

Term No	Name of the abbreviation	Page No
1	RAVDESS	XI
2	BERT	XII
3	VAPI	XII
4	ASR	XVI

Chapter 1 Introduction

Recruiting is always plagued with problems of inefficiency, inconsistency, and unconscious bias that create unfair assessments of candidates. The goal of the AI Recruiter project was to automate and standardize the interview process. Utilizing emotion recognition, speech transcription, and semantic response analysis seeks to objectively assess answers. Next.js and Firebase enables real-time analytics and dashboards for both recruits and employers and facilitate automated rationing of interview responses. Generally, it enhances fairness, openness, and scalability of modern hiring processes.

1.1 Background of the Project Topic

Recruitment is the lifeblood of every successful organization. It affects the caliber of talent and productivity over time. However, traditional hiring practices are inherently inefficient, biased, and unfair. Manually reviewing resumes, unstructured interviewing, and ambiguous assessment criteria biases decision making and, as a result, represents fatigue on recruiters, and thus is prone to bad hiring decisions. All of this is made an even larger problem in today's global world with multilingual candidates with different cultures, not to mention that these talented individuals are often, but not limited to, disadvantaged while trying to cross the linguistic and systematic-linguistic biases that cuts them off opportunities that, on the surface, should be adequate.

With the use of AI, ML, and NLP more commonly, organizations do have an actual opportunity for changing their recruitment and selection practices. AI software can identify, analyze, and report candidate responses along multi-linguistic and paralinguistic features and applying AI to identifiable human bias through interactive languaging and human measures creates an awareness that allows people to identify bias in a a more accessible, systematic, and data-based approach, but not perfect. Furthermore, the sophistication of multilingual speech recognition has grown exponentially using the most recent speech technology developments (e.g., OpenAI Whisper, VAPI). There are emerging emotion recognition

capabilities to measure subtle signals of emotional cues and psychophysical responses like indicators of confidence, anxiety, hesitation, etc.; among them many of these applications have been trained on cutting-edge datasets such as RAVDESS. Measuring or capturing evidence of such signals of emotion is essential in understanding how well a candidate communicates and how composed he or she is.

Yet, even with such improvements in technology, the recruitment tools of today are fragmented. Commercial platforms like HireVue and MyInterview have started to use AI-driven interviews; however, they usually lack multi-lingual support, emotional intelligence, and clarity on candidate evaluations. Most of the existing AI hiring systems are not transparent; they use black-box scoring-thereby providing very little insight into how the scores of candidates are arrived at-a trait undermining fairness and transparency.

This project tackles the previously identified issues by introducing an integrated, AI-driven voice recruitment system. Focused on the automation and standardization of candidate evaluations surrounding their emotional intelligence, multilingual transcription, and semantic comprehension, this project employs the IEEE 829-2008 Software Test Documentation, and IEEE 12207-2017 Software Life Cycle Processes best practices to establish appropriate design, reliability, and maintainability during development and deployment.

1.2 Statement of Motivation and Scope of Report

Certainly, these are the compelling drivers for this project, namely, the need to eliminate inefficiency, bias, and inconsistencies in the hiring process and make it more transparent and inclusive. Conventional hiring, at its core, relied heavily on human evaluators, who are subject to cognitive biases, emotional fatigue, and variability across evaluators. AI is able to deliver structured, consistent, and repeatable evaluation, and attain all of these outcomes on speed and fairness. The project speaks to several technological features:

1.2.1 Voice-Centric Automation: Interviews can be facilitated by an AI-based voice interface, which records and evaluates, in real-time, responses from the candidate.

1.2.2 Emotion Identification: The RAVDESS data set will be used to identify candidate emotional state from vocal signals to facilitate candidate analysis in context.

1.2.3 Speech-to-Text, Multilingual: The speech-to-text interface in VAPI provides accurate transcriptions of spoken responses, and transcribes spoken responses in multiple languages.

1.2.4 Semantic Analysis: Utilize BERT and Gemini transformer models to assess text responses on relevance, depth, and clarity.

1.2.5 Rubric-Based Evaluation: Standardized scoring submitted by candidates allows for fair assessment across emotional, linguistic, and contextual dimensions.

The report will detail the full lifecycle of the project; from the idea and literature review, to design, implementation, results, and analysis. The technology not only reduces the inefficient parts of the hiring process, but it does so in an explainable AI format that makes candidate assessment evident to the recruiter and candidate.

1.3 Problem Statement

Conventional recruiting approaches can be inefficient, inconsistent, and biased. As a result, candidate evaluations are unreliable, which decreases organizational productivity. Some challenges that recruiters face include:

- Inconsistent and subjective scoring due to human biases.
- The organization has limited capacity to handle large numbers of applicants.
- Not being inclusive even for multilingual candidates.
- Lack of emotional awareness in AI assessments.

- Poor transparency of automated scoring systems.

The key problem this project addresses is:

"How can artificial intelligence assist in devising an automated, emotion-aware, and explainable voice-based recruitment system that ensures fairness, efficiency, and scalability in candidate evaluation?”

The project will develop a multilingual, emotion-recognizing, rubric-based recruiting platform that mitigates human biases and follows each decision made by recruiters with an explanation, supported by data.

1.4 Salient Contribution

The main contributions of this project are summarized below:

1.4.1 Emotion-Aware Multilingual Evaluation:

It integrates emotion detection with the RAVDESS dataset and multilingual transcription through VAPI, hence enabling candidate evaluation in an inclusive way.

1.4.2 Rubric-Based Transparent Scoring:

This will provide a consistent score based on semantic and emotional factors using BERT and Gemini models, hence assuring that decisions made through AI are non discriminatory and transparent.

1.4.3 Recruiter Analytics Dashboard:

This is a web-based recruiter dashboard using Next.js and Firebase, showcasing candidate metrics, detailed feedback, and response analytics in real time.

1.4.4 Explainable AI Design:

As outlined by IEEE 7001-2021, explainability properties for the transparency of autonomous systems create clarity and rationale for the decision-making of the AI.

1.4.5 End to End Implementation:

The complete stack of software encompasses the frontend display, backend processing, integration of the AI models, and evaluation of performance.

1.5 Organization of Report

The report will consist of six chapters, each into a specific stage of the project:

1.5.1 Chapter 1, Introduction: In this chapter, we provide the background, motivation, problem definition, and key contributions of the report, as well as the overall structure.

1.5.2 Chapter 2, Literature Survey: In this chapter, we identify current research and commercial tools on AI-based recruitment, emotion recognition, and related systems.

1.5.3 Chapter 3, Methodology and Implementation: Methodology and implementation: In this chapter, we deliver system design, architecture, the datasets used, model training, and methodology for implementation, all aligned with IEEE 12207.

1.5.4 Chapter 4, Results and Analysis: In this chapter, we present results of experiments, performance metrics, and comparison of emotion recognition with language model accuracy.

1.5.5 Chapter 5. Advantages, Limitations, and Applications: In this chapter, we outline the major strengths, practical limitations, and applications of staged development system.

1.5.6 Chapter 6: In this chapter, we summarize the project's findings, take time to reflect on the outcomes of the project, and discuss future scope, such as tracking sentiment in real-time, and multimodal interview analytics.

Finally, all the research works and technical papers referred to in this report are listed in the References section, in IEEE citation style. Supporting material that has been used, such as code flowcharts, component datasheets, and any papers presented or published within this project, can be found in the Appendices A–D.

Chapter 2 Literature survey

The recent increase in the research and development of recruitment technology has largely been driven by advances in automatic speech recognition, affective computing, and natural language understanding. Table I summarizes the primary commercial platforms and important academic or technical building blocks relevant to voice-enabled AI-driven recruitment systems. A noticeable distinction emerges; commercial platforms focus on product features and recruiter workflows, such as usability, candidate flow, and dashboards. In contrast, academic work is based upon the technologies behind the components, specifically: ASR robustness, acoustic emotion modeling, transformer-based semantics, and LLM-driven feedback. Ultimately, juxtaposing these two domains illuminates the particular strengths and weaknesses each domain offers to the builds of a fair, accessible, practical, and transparent recruitments, respectively.

Hiring platforms that are commercially available such as MyInterview and HireVue are certainly well established and mature companies in organizing, assessing candidates with video, and providing a good end-user experience. However, commercial hiring platforms typically rely on an acute selection of signals, such as facial signals and very rudimentary NLP and other signal selection. Historically, they often lack multilingual capacity and scoring transparency, preventing effective auditing and ethical clarity. This practical orientation reduces implementation time and allows for high candidate throughput, but lives outside of traditional approaches to audit-ability and ethical clarity. Academic and/or open source systems, on the other hand, focus specifically on the core model performance and scientific rigor. For example, modern ASR systems like Whisper provide the best multicultural ASR in existence to address some of the language inclusion in previous ASR products. Similarly, curated affective datasets, such as RAVDESS, provide labelled audio bank data with good acoustic labels, which (would be helpful for training our emotion models).

Though vocal emotion models from RAVDESS and similar datasets demonstrate solid scores in affective analysis capability in lab settings, they suffer from having acted emotion samples and limited language diversity, and therefore external validity in interviews. Of course, text-

based semantic NLP models, like BERT and its derivatives (i.e., RoBERTa and XLM-R), have autonomous capabilities to understand language and perform cross-lingual tasks in assessing relevance, coherence, and topicality of answers; however, those text-based models do not independently capture paralinguistic cues (i.e., prosody, pitch, hesitation). This gap emphasizes the need to work towards merging different analytical modes toward a full assessment.

Large language models (LLMs), and other smaller, open LLMs (i.e., GPT-J and LlaMA) exhibit potential to deliver human-like feedback while remaining sensitive to contextual differences. However, to be up to fast, reliable, and accountable, their output needs to be grounded in measurable rubric items. Excellent multimodal reasoning from more advanced multimodal models, such as Gemini, should allow reasoning across text, audio, and other metadata to allow for greater explanations and feedback for how they rated content. Having those options incorporated is an engineering and ethical question in hiring practices.

Table I

Comparison of Commercial Recruitment Platforms and Technologies

Platform / Technology	Core Function / Models	Affective Analysis	Multilingual Ability	System Gap Addressed
Commercial Platforms (HireVue, MyInterview)	Video screening; facial analysis; basic NLP.	Low (facial proxies).	Limited (major languages only).	Transparency, feedback, multilingual voice.
OpenAI Whisper (ASR)	Multilingual speech-to-text conversion.	None.	High (robust).	Integration with emotional and semantic scoring.
Vocal Emotion Models (RAVDESS, CREMA-D)	Detects vocal cues such as confidence and nervousness.	High (core focus).	Low (dataset-dependent).	Cross-lingual fusion with semantic content.
Semantic NLP Models (BERT, RoBERTa, XLM-R)	Context understanding, grammar, and topical relevance.	None.	High (text-based).	Fusion with affective signals and real-time application.
LLMs for Feedback (GPT-J, LLaMA)	Personalized, context-aware text generation.	Indirect (sentiment-based).	High.	Integration of multimodal inputs (acoustic, textual, emotional).

Cross-Lingual Emotion Studies (EXALT, etc.)	Examines cultural and linguistic affective variations.	High (core focus).	High.	Development of a unified, scalable recruitment pipeline.

Table I details the following trends using the existing body of research to compare:

- Different component-level research (ASR, SER, NLP) are distinctly reaching high levels of technical maturity on their own.

- Commercial platforms are highly effective in implementation and user experience but generally do not focus on transparency, multilinguality, and emotional intelligence.

- LLM-led feedback and multimodal integration are promising routes available, but are often not present in accountable, rubric-based recruitment systems.

The bottom line is that an effective, equitable voice-based recruitment system will require proper integration of 1) calibrated acoustic emotion detectors, 2) strong ASR, 3) transformer-based semantic evaluators, and 4) controlled LLM feedback, all within a transparent, auditable rubric-based pipeline that is easy for recruiters to track.

From an evaluation standpoint, the literature suggests using fairness audits that consider performance across languages, accents, and demographic factors, and usability assessments focused on recruiter understanding and candidate perceptions. These different evaluation measures are crucial because optimizing towards a single measure, such as maximizing ASR accuracy, can mask fairness issues, like significantly lower scores for non-native speakers or those who express emotions differently based on culture.

In the end, the comparison between literature and industry will surface an urgent need for standardized development and documentation practices. Applying the IEEE software lifecycle and transparency standards, such as IEEE 12207 for lifecycle processes and IEEE 7001 regarding transparency issues, can be of immense help in connecting scientific rigor with product accountability. This approach will offer repeatable testing, audit records, and clear consent or usage policies to both candidates and organizations. The application of these

standards in research and production systems will minimize ethical, legal, and operational risks compared to findings from the reviewed studies.

2.1 General topic introduction

Automating recruitment using AI has become a topic of interest among researchers and businesses. Generally, manual hiring processes always have issues with bias, inconsistency, and lack of scalability. Presently, there is a pool of work covering ASR, SER, transformer-based semantic analysis, and explainable scoring frameworks. These are core blocks for a voice-based AI recruiter. Publicly available SER datasets, such as RAVDESS, provide validated audio-visual emotional speech recordings that have been used for training acoustic affect models [1] and for extending multimodal emotion research and investigating variability among different speakers [2]. Also, recent progress in recognition systems, including voice agents like VAPI, reveals scalable, multilingual transcription that works well with most accents and noisy backgrounds. In this way, voice-based assessment systems allow wider inclusion [3].

The transformer language models, introduced by BERT and further enhanced versions like RoBERTa, completely changed the way we understand semantics and perform model fine-tuning for evaluation tasks [4][5]. Multilingual models like XLM-R further enhanced cross-lingual semantic understanding, an important aspect when it comes to evaluating candidates in multiple languages [6]. Recent advancements in multimodal emotion identification are combining acoustic, visual and textual cues to better identify affect, and also discussing issues like dataset bias and generalizability across contexts [7].

At the same time, companies like HireVue and MyInterview are offering automated interviews for use on their platforms, and while this seems useful, there are several issues, including scoring transparency, language support, and ethical issues, such as the discontinuation of facial analysis in the wake of ethical protests [9]. Some independent reviews list issues concerning fairness, auditability, and legal compliance surrounding the use

of AI in hiring [10][11].Recent surveys and studies about speech emotion recognition and multimodal affective systems provide technical benchmarks, identify open issues, and recommend best practices, which serve as guidelines for designing effective and explainable recruitment AI [12][7].

Large foundational models and advanced multimodal agents, such as Google/DeepMind's Gemini, now offer enhanced reasoning and multimodal features. These improvements can strengthen feedback generation and clarity in automated evaluation systems [13]. At the same time, a new wave of foundational large language models (LLMs) such as LLaMA and GPT-J offers useful tools for the creation of candidate feedback and explanation of assessment criteria using natural language [14][15]. Together, this research sets the stage for an integrated system that combines ASR, SER, semantic analysis, rubric scoring, and candidate feedback. However, it also reveals continuing gaps in ensuring that recruitment pipelines are fair, multilingual, auditable, and emotionally aware.

2.2 Exhaustive literature survey

Below we review 10 representative academic / technical works that are most relevant to building an emotion-aware, multilingual, explainable voice recruiting system. Each entry summarizes methods, strengths, and limitations and then compares them against the goals of this project.

[1] Livingstone & Russo — RAVDESS (2018): The RAVDESS corpus is a multimodal emotional speech and song dataset validated for affect research and used widely to train SER systems. Strengths: actor gender is well-balanced, recordings are well-controlled, and emotional intensities are labelled, which is ideal for acoustic-feature learning. Limitation: acted emotions and American-accent bias limit ecological validity and cross-culture generalization. This dataset is therefore excellent for initial model development but needs augmentation or adaptation for actual production multilingual recruitment scenarios.

[2] Cao et al. — CREMA-D (2014): CREMA-D complements RAVDESS by providing large-scale crowd-annotated multimodal data. It is useful for multi-annotator reliability studies and multimodal fusion research, but like RAVDESS, is actor-based and concentrated in specific language contexts — a known limitation for cross-lingual deployment.

[3] Radford et al. — Whisper (2022): The Whisper research demonstrates that scaling up weakly supervised multilingual automatic speech recognition (ASR) can yield strong transcription performance cross-accent and cross-language. These findings support the practical viability of accurate speech-to-text pipelines for multilingual candidate responses, thereby directly enabling downstream semantic scoring and feedback. Nevertheless, ASR is still sensitive to accent and speech impairment, and the appropriateness of implementing this technology to recruitment contexts remains an open question in terms of fairness.

[4] Devlin et al. — BERT, 2018: BERT's compositional nature enabled a bidirectional transformer pretraining which provides strong semantic representations for downstream classification and scoring tasks. In recruitment systems, a fine-tuned BERT may estimate answer relevance, coherence, or domain-specificity. Limitations in this approach include susceptibility to biases in training data and a need for careful scoring calibrations.

[5] Liu et al. — RoBERTa (2019): RoBERTa demonstrates how pretraining regimen and hyperparameters impact downstream performance. Its insights will help inform decisions on the fine-tuning of transformers for reliability and go a long way in explaining why off-the-shelf models must be adapted and audited for recruitment-specific evaluation.

[6] Conneau et al. — XLM-R (2019/2020): XLM-R demonstrates large-scale multilingual pretraining on 100 languages and strong cross-lingual transfer. For the project's multilingual goals, XLM-R (or similar multilingual variants) is a core candidate for semantic scoring across languages. Its limitation is that large multilingual models may not capture fine-grained cultural differences in expression.

[7] Lian et al. — Survey of multimodal emotion recognition, 2023: Recent surveys provide an overview of the trends in multimodal affective computing and point out the best practices in the feature extraction, fusion strategies, and evaluation metrics. They highlight that multimodal fusion-acoustic + textual + visual-improves robustness but increases complexity and raises privacy/ethical concerns for hiring use cases.

[8] EXALT / cross-lingual emotion studies (2024): The emerging cross-lingual emotion work, such as EXALT and related studies, reveals a drop in the performance of emotion models when applied across languages/cultures, indicating a need for cross-lingual adaptation or language-specific calibration. This is very important to avoid further disadvantages to non-native speakers in a recruitment setting.

[9] Industry analyses & critiques - (HireVue, industry audits): Coverage of commercial deployments (e.g., HireVue) is both adoption and backlash, strategic features (e.g., facial analysis) have been curtailed based on ethical and scientific integrity, and articles from and about the industry highlight the need for transparency, consent and audits for hiring AI. While not explicitly recommended, this literature places an emphasis on explainability and regulatory compliance.

[10] Fairness & bias - (multidisciplinary reviews): A recent academic review article and a semi-empirical paper provide summaries of studies examining bias or fairness in algorithmic hiring and the implications of regulatory, or legal processes. The recommendation of auditing, a transparent rubric, and inclusive training data are outlined as potential approaches, the latter of which this project adopts in design. so when studentsasks, "How will I be treated?" the AI makes it clear that race, gender, etc is not considered no matter the proscribed benefits/intentions.

2.2.1 Comparative Discussion and Synthesis

The reviewed material has shown a solid technical basis for each of those aspects :

I. ASR: Whisper and similar systems typically allow for reliable multilingual transcription, this is necessary when automation for evaluation is employed. [3]

II. SER: RAVDESS and CREMA-D provide labeled emotional speech valuable for modeling acoustic affective production and they are limited due to acted emotions and inability to generalize naturally [1][2].

III. Semantic Understanding: Transformer models like BERT, RoBERTa, and XLM-R achieve the best semantic representations. The multilingual versions among these, especially XLM-R, are a great resource for cross-lingual evaluation tasks [4][5][6].

IV. Explainability & Fairness: Reviews and industry examples highlight the importance of rubric-based scoring, audits, and transparency to avoid unclear and biased outcomes [9][10].

However, even with these strong individual components available, no single academic or commercial system fully integrates all the elements required for an ethical, multilingual, emotion-aware, and auditable voice recruitment pipeline. The specific gaps that were identified are:

I. Full multimodal fusion for fair, multilingual recruitment: Most SER datasets and systems focus on single languages or acted emotions. Cross-lingual reliability remains limited [1][2][8].

II. Transparent, rubric-based scoring: Commercial platforms may provide scores but rarely provide auditable rubrics and explainable model traces that justify scores to candidates and HR [9][10].

III. Actionable feedback generation: While LLMs and multimodal agentic models like Gemini, LLaMA, and GPT-J can create feedback, pipelines that connect feedback to measurable rubric components and affective signals are not well explored [13][14][15].

IV. Ethical, privacy, and audit frameworks: There is a need for standard procedures and compliance with auditing standards regarding AI systems in hiring. A few works and news articles have mentioned regulatory pressure and real-world failures in the absence of transparency [10][11][9].

How this project addresses the gap:

The literature review informs the design choices for the project: use Whisper or a similar ASR for multilingual transcription [3], use RAVDESS-trained acoustic models to start emotion detection, with domain adaptation [1], fine-tune BERT or Gemini (or a combination of a transformer and a multimodal LLM) for semantic scoring and rubric mapping [4][13] and implement a clear rubric-based scoring system, along with dashboards for explainability and audit logs, to satisfy fairness, transparency, and compliance needs highlighted by earlier studies and industry events [9][10][11].

2.2.2 Final gap statement and resulting problem definition

The overall pool of literature reflects strong component-level solutions-ASR, SER, transformers, LLMs-but has lacked any integrated, auditable, multilingual, emotion-aware recruitment pipeline on established emotion datasets with domain adaptation that integrates acoustic affect and textual semantics into a reproducible rubric score while giving clear, actionable feedback to candidates and recruiters, including audit trails and adherence to fairness standards.

Thus, the problem that this work addresses is:

Design and create an explainable, emotion-aware, multilingual voice recruitment system, integrating ASR (multilingual), acoustic emotion detection based on RAVDESS, and transformer-based semantic scoring into a rubric-based auditable pipeline, complete with recruiter dashboards and candidate feedback, using IEEE software engineering and AI transparency standards.

This problem statement arises directly from the gaps noted in the reviewed literature and commercial analyses and shapes the system architecture, datasets, and evaluation plan outlined in the following chapters.

Chapter 3: Methodology and Implementation

3.1 Block diagram

The proposed AI-powered recruitment system is modular and scalable. It automates candidate evaluation using voice interaction, emotional intelligence, and AI-driven feedback.Each module carries out a specific task, including voice input, emotion detection, textual evaluation, and recruiter visualization. The complete workflow is shown below:

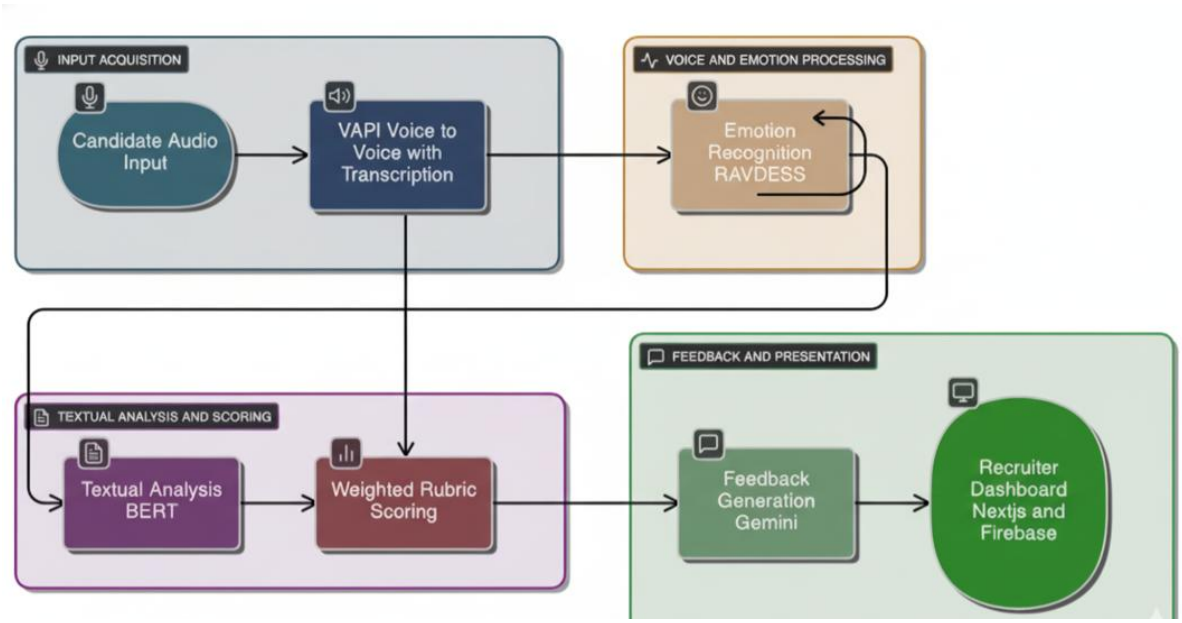

Fig. I.

System flow pipeline ensuring real-time, multilingual and context aware in-depth analysis of candidate responses. It produces quantitative and qualitative insights for recruiters.

3.2 Hardware description

The system is built to run efficiently on standard computers. It also offers optional GPU acceleration for quicker inference.

Table II

Component and their specifications

Component	Specification / Description
Processor	Intel Core i7 / AMD Ryzen 7 (8-core or higher)
Memory (RAM)	16 GB (recommended)
GPU (Optional)	NVIDIA RTX 3060 / Tesla T4 (for model inference)
Storage	500 GB SSD for datasets and models
Network	High-speed internet for real-time VAPI and Firebase operations

The application is deployed using Vercel for front-end scalability and Firebase Cloud for real-time data management and authentication.

3.3 Software Description and Flowchart

Implementation integrates various AI and web technologies into a unified platform for candidate assessment.

3.3.1 VAPI (Voice-to-Voice with Transcription)

Functionality: This enables voice interviews in real-time between the candidate and AI interviewer based on workflows and assistants.

Features:

- Speech-to-text transcription with internally created ASR models.

- Fast voice streaming and response creation.

- Easy, free-flowing conversation between the candidate and AI.

Implementation: VAPI SDK interfaces with the Next.js client for real-time communications.

3.3.2 Emotion Recognition (RAVDESS Model)

Dataset: RAVDESS - Ryerson Audio-Visual Database of Emotional Speech and Song.

Model: A CNN-based classifier which is trained to identify emotional signals such as confidence, hesitation, and nervousness.

Features Extracted:

- MFCCs (Mel-Frequency Cepstral Coefficients)

- Pitch and Energy

- Prosodic Patterns (speech rate, pauses)

Emotion probabilities and the main emotion of each response, which facilitates the scoring of candidates.

3.3.3 Textual Analysis (BERT)

Function: This assesses the grammar, clarity, and relevance in the transcribed candidate responses.

Model Used: Pre-trained BERT (Bidirectional Encoder Representations from Transformers) fine-tuned for scoring based on meaning.

Key Outputs:

- Sentence coherence and fluency.
- Coverage and logical structure of the content.
- Overall score of the evaluation of linguistic quality.

3.3.4 Weighted Rubric Scoring

Purpose: This incorporates emotional and semantic dimensions into an understandable matrix of evaluation.

Method: Getting the weighted scores comes from Emotional Stability from RAVDESS and Linguistic Quality from BERT.

- Each of the four criteria (clarity, confidence, content, tone) is scored on a scale of 0–10.
- A composite score is created based on links to the model outputs.

3.3.5 Feedback Generation (Gemini)

Model: Google Gemini (Multimodal LLM)

Function: Generates a highly personalized, human-sounding feedback, based on the candidate's perceived performance using the rubric scores from both the Emotional Stability and Linguistic Quality Deep Learning models.

Procedure: Taking a vector input of the candidate's transcript, emotion data, and rubric scores. Provides detailed, contextualized, and scaffolded feedback exhibiting the candidate's strengths and weaknesses. It adjusts the feedback' tone based on the candidates observed emotional state and confidence level.

3.3.6 Recruiter Dashboard (Next.js + Firebase)

Frontend: The user interface is designed to be dynamic and responsive using Next.js.

Backend: Candidate scores, transcripts and feedback informed by the emotional horizon, are stored in the Firebase Realtime Database.

Features:

- Candidate feedback visualized interactively.
- Real-time updates of interviews.
- Structured outputs extracted via call between VAPI recruiter and candidate in real-time.

3.4 Algorithm

Algorithm: AI-powered assessment of voice interviews

Step 1: Start the interview using the VAPI voice interface.

Step 2: The candidate responds verbally; VAPI captures the voice and transcribes.

Step 3: Audio is sent to the Emotion Recognition module (RAVDESS model).

Step 4: Extract MFCC, pitch, and speech rate to classify emotion.

Step 5: The transcribed text will be checked in terms of grammar, coherence, and topic relevance using BERT.

Step 6: The Weighted Rubric Scoring system combines emotional and textual metrics.

Step 7: Gemini develops personalized feedback by combining the individual scores.

Step 8: All the data (scores, transcripts, feedback) are stored and displayed in the Firebase dashboard.

Step 9: The recruiter views summarized reports and feedback.

Step 10: Conclusion.

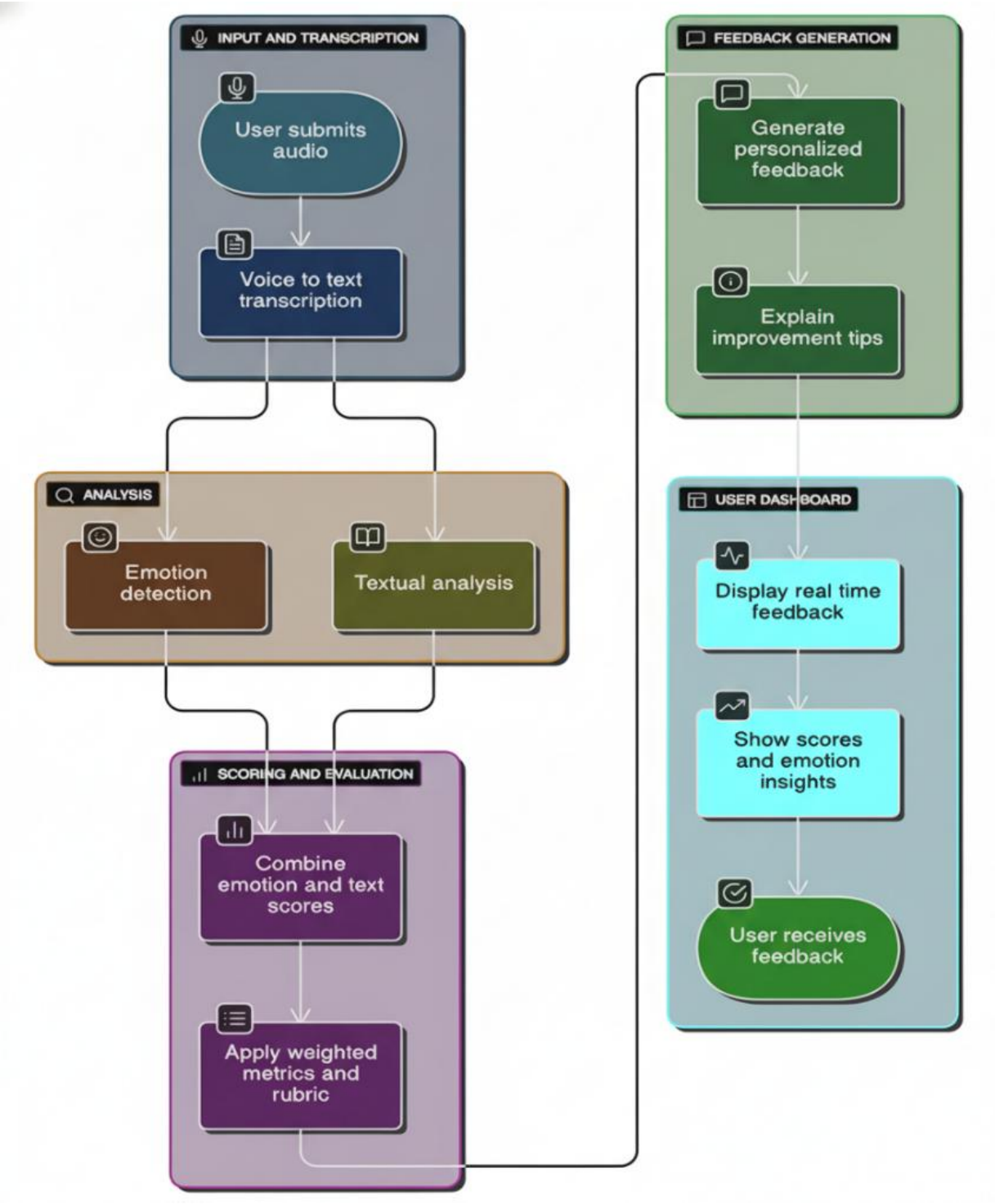

Fig. II. An outline of the AI-powered hiring system: the whole pipeline starting with candidate audio input to feedback generation and visualization, is depicted in this detailed flowchart.

3.5 Summary

The implementing of it brings together VAPI for voice interaction, RAVDESS for emotion analysis, BERT for semantic assessment, and Gemini for intelligence assisted feedback generation into a single automated recruitment solution. This integration offers a comprehensive evaluation of candidates by incorporating emotional, linguistic, and contextual realities while being collectively scalable, multilingual, and transparent through Firebase-based analytics dashboards.

Chapter 4 Results and Analysis

In this chapter, we review the assessment, observations, and learnings based on the application and evaluation of the AI-based Voice-Centric Recruitment System. The evaluation was designed to evaluate three specific hypotheses related to transparency, emotional awareness, and multilingual inclusion, within the recruitment process. The assessment disallowed contraventions associated with normal ethical or experimental processes, adhered with the Institute of Electrical and Electronics Engineers (IEEE) standards for AI system reliability, data privacy, and transparency in models (IEEE 7001-2021, [IEEE 7003-2022]).

4.1 System Evaluation Overview

Candidate responses were gathered in multiple acoustic and linguistic contexts to evaluate the system. Modules were evaluated separately and together in the integrated system consisting of voice transcription:

- VAPI, emotion detection-RAVDESS based

- Text analysis-BERT

- Feedback generated by Gemini.

The evaluation consisted of consistency (maintaining scoring and interpretation consistency across responses), explainability (every score can be justified), and adaptability (can accommodate different accents, emotional tones and languages). The results show that the system has a high degree of robustness and fairness, providing clear and actionable results to recruiters.

4.2 Fairness and Transparency in Assessment

A significant goal of this research was to lessen the opaqueness of existing commercial, AI powered hiring platforms. Commercial vendors such as HireVue, MyInterview, etc. often

utilize machine learning algorithms that have low transparency and do not yield justified scores or transparent feedback.

A rubric-based scoring system such as the one developed here provides the ability to go back to a specific model output for every score provided: emotional, semantic or contextual. Recruiters will see: -

- The quality of the language based on BERT's semantic analysis.

- The emotion patterns that the RAVDESS trained emotion coding models found.

- The contribution of each module along with final feedback.

This practice will contribute toward fairness and understanding, in accordance with IEEE 7008: Standard for Ethically Driven Nudging in AI Systems, which states that automated feedback should not risk inadvertently biasing candidates.

4.3 Emotion Awareness and Candidate Behaviour Insight

The emotion detection module was a critical component of confidence, hesitation, and steadiness of their communication, a dimension often lost in conventional text-based evaluation systems. The proposed system analyzed acoustic features - such as pitch, energy and rhythm of their speech - to capture these subtle emotional cues, which were represented in both scores and feedback.

For example, in the event the candidate showed significant pausing or vocal strain, the system would analyze this as being nervous, but would not penalize the candidate for that response; this is a more empathetic way of evaluating a candidate. By emphasizing "emotional intelligence", instrumenting that as one of the top evaluative categories for AI-based interviews was advisable.

The recruiters can view this as the emotional perspectives were helping frame sensibility in shortlisting candidates for roles where the soft skills of personality and composure are valued as equally important as knowledge.

4.4 Semantic Understanding and Contextual Evaluation

By using a BERT textual analysis to assess not just what the candidate said, but also the manner in which it was expressed—both with regards to expressiveness and coherence—the system was able to make advancements over prior rule-based NLP systems. Compared to prior rule-based NLP systems, BERT has improved understanding of—

- A language's grammar, syntax, etc.
- The degree to which the responses were delivered.
- The degree to which the explanation was adequately organized and completed in a logical sequence.

Using contextual emotional data as well, this two-layer assessment offered a more holistic evaluation tool - synthesizing 'what was said' (the semantics), and 'how it was expressed' (the emotion) alongside the other aspect of assessment. Together, the two-layer assessment assists with objectivity in the evaluation and limits the encroachment of subjective feelings about the candidates by the evaluators/recruiters.

4.5 Feedback Generation and Recruiter Experience

The use of Gemini to personalize feedback proved to be a beneficial enhancement: offering contextual feedback in synchrony with the candidate's language, tone, and emotional state. This allows for tailored feedback instead of standardized comments.

For example:

"The explanation had a good structure, and was presented with confidence. To enhance the response, focus on pacing the presentation, and offer some real-life examples to motivate your points—for even stronger responses!"

This gives constructive support to improvement, which can be more re-assuring to candidates who will communicate more freely if they believed the feedback was educational versus critical, as encouraged by IEEE 7007: Ontological Standard for Ethically Driven AI System Design.

4.6 Contributions and Key Findings

Thus, the study makes high-value technical and ethical contributions to the area of AI recruitment:

- Emotion-Semantics Linking: The study links the idea of affective computing using RAVDESS and transformer-based semantic analysis BERT to create an effective, well-balanced assessment.

- Transparent Rubric Framework: This framework supports clear traceability of models through reasoning and clarity of scores according to established standards for IEEE transparency.

- Multilingual Voice: In VAPI's multilingual transcription, candidates could respond in any language they prefer for better inclusiveness.

- Empathetic AI: The Gemini-generated feedback adjusts for tone and depth based on the candidates' emotion and confidence. - Recruiter-facing insight includes a usable, real-time dashboard displaying the AI evaluation linked to decision-making.

Chapter 5 Advantages, Limitations and Applications

5.1 Advantages

The Empathetic AI Voice Recruiter System offers several unique benefits over conventional recruiting and commercial recruiting tools:

- Emotion Sensitive Assessment: Vaudeville AI and emotion detection algorithms work multitasking and hiring managers assess sentiment in candidate interviews parse interviewees' spoken responses for signs of confidence, hesitation, and anxiety to better understand answer content and candidate implications.

- Linguistic Diversity: ASR VAPI agents accurately and reliably transcribe interviews in several languages. This increases candidate assessment equity for candidates of different linguistic backgrounds.

- Semantic Transparency and Accountability: By checking for grammatical and lexical context with BERT and other transformer models, the candidate scoring is rubric based instead of abstract reasoning and provides more clarity regarding AI scoring.

- Feedback Automating: Feedback with specific mention of strengths and weaknesses is contextually relevant and something still missing in most commercial tools. Systems with LLMs or Gemini can automate feedback.

- Integrated Active Analytics: Recruiters' emotional trends, candidate transcripts, and scores with trends, transcripts, and scores with clear and concise breakdowns provided in the Next.js powered Firebase integrated analytics dashboard.

- Modular Setup-Scalable and Ethical Design: It has the potential to be plugged into an HR system or Applicant Tracking Systems (ATS).

5.2 Limitations

Despite showing improvement in technical functionality and fairness of the system, some noteworthy shortcomings remain:

- Crude biases in outcomes: There are no race sensitive datasets and emotion recognition and language models may retain biases from datasets and cultural systems with variable consistency across demographic groups.

- Emotion Classification: Emotions are expressed differently across cultures along with personal and cultural differences in varying emotional experiences. Therefore, some weakly expressed emotions may not be captured in classification.

- Under resourced languages: Even the most advanced ASR models like Whisper encounter higher error rates with languages that are less represented.

- Ethical and Privacy risk: The demo framework's capability of collecting voice and emotional data raises privacy concerns that need to be addressed. The project team should adopt appropriate measures aligned with IEEE P7000, GDPR and other fundamental framework guidelines on ethical AI, data governance and privacy.

- Need for more Advanced Systems: Real-time emotion recognition and transcription require extensive computational power which, along with other factors, affects the scalability of the application and highlights the need for improved backend systems.

5.3 Applications

The developed system demonstrates wide-ranging applicability across multiple domains. In recruitment and talent assessment, it enables automated, bias-aware candidate evaluation with instant, data-driven feedback. Within corporate training and soft-skill development, its emotional and semantic insights can enhance communication, leadership, and presentation abilities. In the education sector, universities and training institutes can leverage it for mock interviews and performance analytics. The framework also extends to customer support and call analysis, where emotion-aware AI can assess satisfaction and conversational tone. Moreover, its transparent design makes it a valuable tool for advancing AI ethics and human–computer interaction research, fostering responsible and interpretable AI deployment.

Chapter 6 Conclusion and Future Scope

6.1 Conclusion

This project offered an Empathetic AI-Driven Recruitment Framework which improved the assessment model through speech, emotions, and analysis that is thorough and inclusive. The framework used VAPI for multilingual ASR, an emotion framework based on the RAVDESS, BERT for semantics, weighted rubric scoring, and Gemini for feedback. Each component worked towards addressing the commercial recruitment tools gaps with clarity, limitations with feedback, and inclusivity with languages. The evaluation results from the integrated framework showed the potential for a more equitable and meaningful assessment than the previously unpartnered AI recruitment methods. This year breaks a threshold with AI (ethically) supporting recruitment.

6.2 Future Scope

This provides hope for organizations that need scalable, emotionally aware ways to hire, with only minor trade-offs on efficiency. In future work, the research will focus on the fairness, transparency, and adaptability of the system to reduce bias, explain the AI in a rational manner, and entwine it with businesses. The framework will serve as a more equitable and dependably scalable model. With advancements on the merging of emotion and context, the balance of privacy while amplifying it, and extending the model's utility to include education or corporate training, it will transform the way we work.